**Craig M. Nielsen** | New York University, College of Arts and Sciences
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## EDUCATION

**New York University** | College of Arts and Science | New York, NY Sep 2023-May 2027
Honors Bachelor of Science | Data Science | GPA 3.95
Minors: Mathematics, Economics, Music
**Relevant Coursework:** Honors Real Analysis I, Linear Algebra and Optimization for Machine Learning, Fundamentals of Machine Learning, Calculus III, Probability & Statistics, Causal Inference, Data Management and Analysis, Intermediate Microeconomics, Linear Algebra.
**Involvement:** NYU All-University Jazz Orchestra Bassist, Music Mentors Collaborative Instructor, NYU Orchestra Bassist, NYU Prague Blog

## HONORS AND AWARDS

Awarded $1250 from NYU Dean's Undergraduate Research Fund for research on *Why People Listen to Sad Music*.
Applying for $1250 from Dean's Undergraduate Research Fund for research on *Mapping Timbre Space*.
Awarded a fully funded study abroad opportunity through NYU's Dean's Presidential Honors Society
Nominated to submit an original essay composition to the Mercer Street publication by Professor William Morgan (NYU).

## RESEARCH EXPERIENCE

**Research Assistant** | *Fox Lab NYU* | New York, NY Jan 2026-Present
- Meet weekly with Dr. Pascal Wallisch, Department of Data Science, to discuss research and analyze data.
- Running participants weekly for a study on pattern recognition in chess
- Awarded a $1250 grant on Why People Listen to Sad Music

**Research Assistant** | *Dr. Laurence Maloney's Lab NYU* | New York, NY July 2026-Present
- Meet weekly with Dr. Laurence Maloney, Department of Psychology, to discuss experiment design and grant writing
- Designing a study investigating the geometry of timbre perception
- Applying for a $1250 grant on Mapping Timbre Space

**Research Assistant** | *CROWE University of Wisconsin-Madison* | Madison, WI June-July 2025
- Worked with Dr. Ananth Seshadri, Department of Economics, and met with a post-doc researcher weekly
- Created a model measuring the causal effect of a tax credit on manufacturing employment
- Created data visualizations depicting said tax credit's effect
- My work contributed to the publication of this report

## RESEARCH INTERESTS

Domain shifts, high-dimensional statistics, machine learning for the social sciences.

## PROJECTS

**Michelin Star Restaurant App** | *NYU* | New York, NY Jan-May 2026
Developed in Fundamentals of Machine Learning course with Kyunghyun Cho, PhD.
I am entirely responsible for the recommendation engine of the app.
Domain-shifted large Yelp data to sparse Michelin star restaurant reviews in NYC using the Python library Hugging Face. The model estimated rating distributions with a multi-headed Laplace distribution, using a corruption function to focus the model on target restaurants.

**Effect of Corporate Income Tax on Manufacturing Employment** June-July 2025
*CROWE University of Wisconsin-Madison* | Madison, WI
Created a difference-in-differences model on the effect of corporate income tax on manufacturing employment in Wisconsin and bordering states.
Received and aggregated data from the Quarterly Census of Employment and Wages.
Cleaned and analyzed data using Pandas and Matplotlib in Python.

**ServiceNow AI** | Spectrum Brands | Middleton, WI May-Aug 2024
Created an analysis of business costs and benefits for implementing ServiceNow AI at a Fortune 500 company.
Presented to company leaders, including the CTO, with a team of 3 other interns.

## TEACHING EXPERIENCE

**Instructor** | *Lavner Education* | New York, NY June-Aug 2026
- Planned and taught introductory computer science and machine learning courses.
- Courses included *Agentic AI and Machine Learning* and *Python & App Development*.
- Taught private lessons.

## OTHER EXPERIENCE

**Desktop Support Intern** | *Spectrum Brands* | Middleton, WI May-Aug 2024
- Duties included responding to service desk calls, resolving ServiceNow tickets through ITIL best practices, and customer service.
- Learned SAP basics through assisting users with SAP-related tickets and meetings with company IT experts.
- Used Azure Device Management to enforce security protocols and assist users.

**Customer Service** | Multiple Companies June 2021-Aug 2023

## SKILLS

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|---|---|
| **Programming Languages:** | Java, JavaScript, Python, SQL, R, MATLAB |
| **Python Packages:** | PyTorch, TensorFlow, Matplotlib, Pandas, NumPy, scikit-learn |
| **Business Applications:** | Power BI, Excel, ServiceNow, MS Office Suite |
| **Languages:** | German, intermediate B2 proficiency |

## PAID MUSICAL PERFORMANCES

La Brioche | Madison, WI July 2024-25
Harmony Bar and Grill | Madison, WI July 2024
Middleton High School Band & Orchestra Parents Association | Middleton, WI May 2023
Dane County Buy Local | Middleton, WI Feb 2023

## REFERENCES

Pascal Wallisch, PhD: Data Science, Neural Science, and Psychology | New York University
212-998-3775 | pascal.wallisch@nyu.edu

Ananth Seshadri, PhD: Economics | University of Wisconsin - Madison
608-262-6196 | ananth.seshadri@wisc.edu

Laurence Maloney, PhD: Psychology, Neural Science | New York University
212–998-7851 | laurence.maloney@nyu.edu